

Trip Simulation Formulas and Fuel Card Feature

Service Tracking MS - Functional Calculation Documentation

Prepared on 01 August 2026

This document summarizes the formulas, validations, settings, and fuel card controls implemented in the Trip Simulation workflow. It is written for operations, finance, procurement, and system users who need to understand how values are calculated and controlled.

Primary source modules reviewed: Trip Simulation server logic, Trip Simulation client logic, Trip Settings, Fuel Card Ledger Entry, Fuel Card Recharge, Fuel Card Issue, and Fuel Card Movement Report.

1. Executive Summary

Trip Simulation is the planning document used to estimate expected revenue, route distance, fuel requirements, fixed route expenses, total trip cost, net profit, and net profit margin before creating commercial and procurement documents.

After a Trip Simulation is submitted, users can create a Quotation and Purchase Orders from it. If a Fuel Card is selected, the system creates a Fuel Card Ledger Entry on submit to deduct the required fuel quantity and value from the card.

Important control: Trip Simulation cannot save/submit successfully when Net Profit Margin is below the Targeted Net Profit percentage.

2. Main Trip Simulation Formulas

Area	Formula	Source / Meaning
Days in Trip	Return Date - Departure Date + 1	Inclusive date count. If return date is before departure date, saving is blocked.
Total Distance (km)	Sum of all Fuel child-table Distance rows	Distances are loaded from Simulation Routes trip steps and stored in the Fuel child table.
Fuel Used (Ltr)	Distance x Truck Type Litres per KM	Each fuel route row calculates its litres using the selected Vehicle truck type setting.
Total Fuel Used (Ltr)	Sum of Fuel Used (Ltr) from all Fuel rows	Stored in total_fuel_consumption_qty_ratio.
Fuel Price	Last purchase rate of Fuel Item	Fetches first from submitted Purchase Invoice Item. If none exists, fetches from submitted Purchase Order Item.
Total Fuel Costs	Total Fuel Used (Ltr) x Fuel Price	This is the monetary fuel requirement for the simulation.
Total Trip Cost	Total Fuel Costs + Sum of Trip Expenses Outline Amount	Combines fuel and all calculated trip expense rows.
Net Profit	Expected Revenue - Total Trip Cost	Stored in net_profit.
Net Profit Margin %	Net Profit / Expected Revenue x 100	Stored in net_profit_. If expected revenue is zero, margin is zero.

Note on the old wording: $100 - ((\text{Expected Revenue} - \text{Total Trip Cost}) / \text{Expected Revenue}) \times 100$ is a cost ratio, not net profit margin. The implemented Net Profit Margin is $\text{Net Profit} / \text{Revenue} \times 100$.

3. Trip Settings Used by Formulas

Setting	Default	Used For
Management Fee Percentage	3%	Management Fee = Expected Revenue x Management Fee % / 100.
Salaries Percentage	10%	Salaries = Expected Revenue x Salaries % / 100.
Heavy Truck Vehicle Cost	85,000,000	Depreciation for vehicles whose truck type is Heavy Truck.
Light Truck Vehicle Cost	45,000,000	Depreciation for vehicles whose truck type is Light Truck.
Heavy / Light Truck Tyre Price	Manual	Used in Tyres formula by truck type.
Heavy / Light Truck Number of Tyres	Manual	Used in Tyres formula by truck type.
Heavy / Light Truck Tyre Lifecycle (km)	Manual	Used to calculate tyre cost per km.
Heavy / Light Truck Litres per KM	Manual	Used to calculate fuel quantity. Example: 0.300 means 0.300 litres per km.

4. Expense Formulas

Expense	Formula	Notes
Management Fee	Expected Revenue x Management Fee % / 100	Percentage comes from Trip Settings. Default is 3%. Quantity stores the percentage; rate is Expected Revenue / 100.
Salaries	Expected Revenue x Salaries % / 100	Percentage comes from Trip Settings. Default is 10%.
Maintenance Fee	Previous Month Maintenance Cost / 30 x Days in Trip	Previous month cost is read from submitted Purchase Invoice Items linked to Purchase Orders that are linked to EAH Job Cards for the selected Vehicle.
Tyres	Tyre Price x Number of Tyres / Tyre Lifecycle KM x Total Distance KM	Tyre settings come from Trip Settings by truck type. Tyres do not fetch amount from Fixed Expenses and are exempt from route amount restriction.
Depreciation	Vehicle Cost / Month Number / 12 / 30 x Days in Trip	Vehicle cost comes from Trip Settings by selected Vehicle truck type. Month Number is the month number of departure/transaction date.
Per Trip Day Expense	Fixed Expense Amount x Days in Trip	Used for route expenses configured with Calculation Method = Per Trip Day.
Salary Allocation Method	Salaries Constant / 30 / Active Vehicles x Days in Trip	Legacy calculation method still supported for expenses configured that way.
Percentage of Expected Revenue	Expected Revenue x Percentage / 100	Percentage can come from Fixed Expenses or the child row quantity where applicable.
Fixed Amount	Fixed Expense Amount	Fallback when no special calculation method applies.

5. Maintenance Fee Source Logic

The Maintenance Fee uses the previous month maintenance cost for the selected Vehicle. The reference date is Departure Date; if Departure Date is empty, Transaction Date is used.

Step	Logic
1	Calculate previous month range from the reference date. Example: if departure is 11-06-2026, previous month is 01-05-2026 to 31-05-2026.
2	Find submitted Purchase Invoice Items.
3	Only include invoice items linked to Purchase Orders.
4	Only include Purchase Orders linked to EAH Job Card using the available job card link field.
5	Only include linked EAH Job Cards where Vehicle equals the selected Trip Simulation Vehicle.
6	Sum Purchase Invoice Item base net amount for that vehicle and previous month.

Maintenance daily rate = Previous Month Maintenance Cost / 30. Maintenance Fee = Maintenance daily rate x Days in Trip.

6. Route and Expense Controls

Control	Behavior
Upsert-style expense handling	Expense names are normalized with trim, lowercase, and spacing cleanup before comparison. This prevents duplicates such as MANAGEMENT FEE and Management Fee.
Standard labels	Driver Mileage, Salaries, Car Wash, Maintenance Fee, Depreciation, Management Fee, and Tyres are canonical labels.
Duplicate prevention	Saving is blocked if the same normalized expense appears more than once in Trip Expenses Outline.
Route restriction	For non-Tyres expenses, row amount cannot exceed the amount calculated from the selected Simulation Route and formula rules.
Tyres exception	Tyres is calculated from Trip Settings and distance, and is not blocked by route predefined amount.
Not Saved prevention	Client recalculation is guarded so submitted/saved documents are not dirtied unnecessarily during refresh.

7. Fuel Card Feature

The Fuel Card feature tracks prepaid fuel balance in both litres and monetary value. It supports recharges, manual issues, automatic trip usage deductions, cancellation reversals, and a movement report.

Document	Purpose
Fuel Card	Master record holding card name, supplier, fuel item, currency, opening litres/value, current balance litres/value, and status.
Fuel Card Recharge	Records litres added to the card. On submit, creates a ledger entry with Litres In.
Fuel Card Issue	Records fuel issued outside Trip Simulation, such as office car fuel or other issuing. On submit, creates a ledger entry with Litres Out.
Fuel Card Ledger Entry	Immutable movement ledger for recharges, trip usage, other issues, and cancellations.
Fuel Card Movement Report	Shows all in/out quantity and value movements, rate, reference, balance quantity, and balance amount after each transaction.

8. Fuel Card Formulas

Calculation	Formula
Ledger Entry Amount	Litres In or Litres Out x Rate
Balance Litres After Transaction	Current Card Litres + Litres In - Litres Out
Balance Value After Transaction	Current Card Value + Amount In - Amount Out
Fuel Card Current Balance Litres	Opening Litres + Sum(Litres In - Litres Out)
Fuel Card Current Balance Value	Opening Value + Sum((Litres In - Litres Out) x Rate)
Trip Usage Deduction	Total Fuel Used (Ltr) deducted as Litres Out at Fuel Price
Recharge Amount	Recharge Litres x Recharge Rate
Issue Amount	Issue Litres x Issue Rate

9. Fuel Card Submission Controls

Control	Behavior
One direction per ledger row	A ledger entry must have either Litres In or Litres Out, never both.
Non-empty movement	A ledger entry must have either Litres In or Litres Out greater than zero.
No negative litres	If a transaction would make card litres negative, the system blocks it.
No negative monetary value	If a transaction would make card value negative, the system blocks it and tells the user to refill the Fuel Card.
No ledger deletion	Fuel Card Ledger Entry cannot be deleted. Users must create a reversing entry through cancellation.
Trip submit deduction	When Trip Simulation is submitted and Fuel Card is selected, the system deducts total fuel used from the card.
Trip cancel reversal	When Trip Simulation is cancelled, the system creates a cancellation ledger row adding the fuel back.
Fuel Price required	Trip usage deduction requires a fuel price greater than zero.

10. Fuel Card Dashboard on Trip Simulation

When a Fuel Card is selected, the Trip Expenses tab shows an availability dashboard. It compares the requested fuel amount and litres against the current card balances.

Displayed Value	Formula / Meaning
Card Amount Balance	Fuel Card current_balance_value
Requested Amount	Total Fuel Costs
Amount After Request	Card Amount Balance - Requested Amount
Card Qty Balance	Fuel Card current_balance_litres
Requested Qty	Total Fuel Used (Ltr)
Qty After Request	Card Qty Balance - Requested Qty
Status	Available when card is selected, not blocked, amount after request >= 0, and qty after request >= 0. Otherwise Refill Needed.

11. Document Creation from Trip Simulation

Action	Rule
Create Quotation	Available after Trip Simulation is submitted. Requires Expected Revenue greater than zero, Customer selected, and Project Sales Item configured. Quotation rate equals Expected Revenue.
Create Purchase Order	Available after submission for payable trip expenses. Each selected payable expense creates a Purchase Order item using row quantity, rate, amount, description, and Fixed Expenses item.
Create Fuel Purchase Order	Available after submission. Requires Fuel Supplier, Fuel Item, Total Fuel Used greater than zero, and Fuel Price greater than zero. PO item quantity equals Total Fuel Used and rate equals Fuel Price.
Linkage	Quotation and Purchase Order store a custom Trip Simulation link where the custom field exists. Trip Simulation stores created Quotation / Purchase Order references.

12. Workspace and Reports

Area	Included Items
Trip Simulation Center Workspace	Number cards for simulations, revenue, cost, profit/loss, fuel balance litres, and fuel balance value.
Workspace Charts	Revenue vs Cost vs Profit/Loss bar chart and Trip Cost Breakdown pie chart.
Trip Analysis Report	Columns include revenue, total trip cost, total fuel cost, fuel litres used, management fee, maintenance fee, tyres cost, total expenses, net profit, net margin, and margin variance.
Fuel Card Movement Report	Columns include posting date, fuel card, movement type, reference, vehicle, driver, in/out qty, rate, amount in/out, balance qty, balance amount, and remarks.

13. Quick Formula Reference

Days in Trip = Return Date - Departure Date + 1

Fuel Used (Ltr) = Distance x Litres per KM

Total Fuel Costs = Total Fuel Used x Fuel Price

Management Fee = Expected Revenue x Management Fee % / 100

Salaries = Expected Revenue x Salaries % / 100

Maintenance Fee = Previous Month Maintenance Cost / 30 x Days in Trip

Tyres = Tyre Price x Number of Tyres / Tyre Lifecycle KM x Total Distance KM

Depreciation = Vehicle Cost / Month Number / 12 / 30 x Days in Trip

Total Trip Cost = Total Fuel Costs + Sum Trip Expenses

Net Profit = Expected Revenue - Total Trip Cost

Net Profit Margin % = Net Profit / Expected Revenue x 100

Fuel Card Amount = Litres x Rate

Fuel Card Value Balance = Opening Value + Sum((Litres In - Litres Out) x Rate)

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